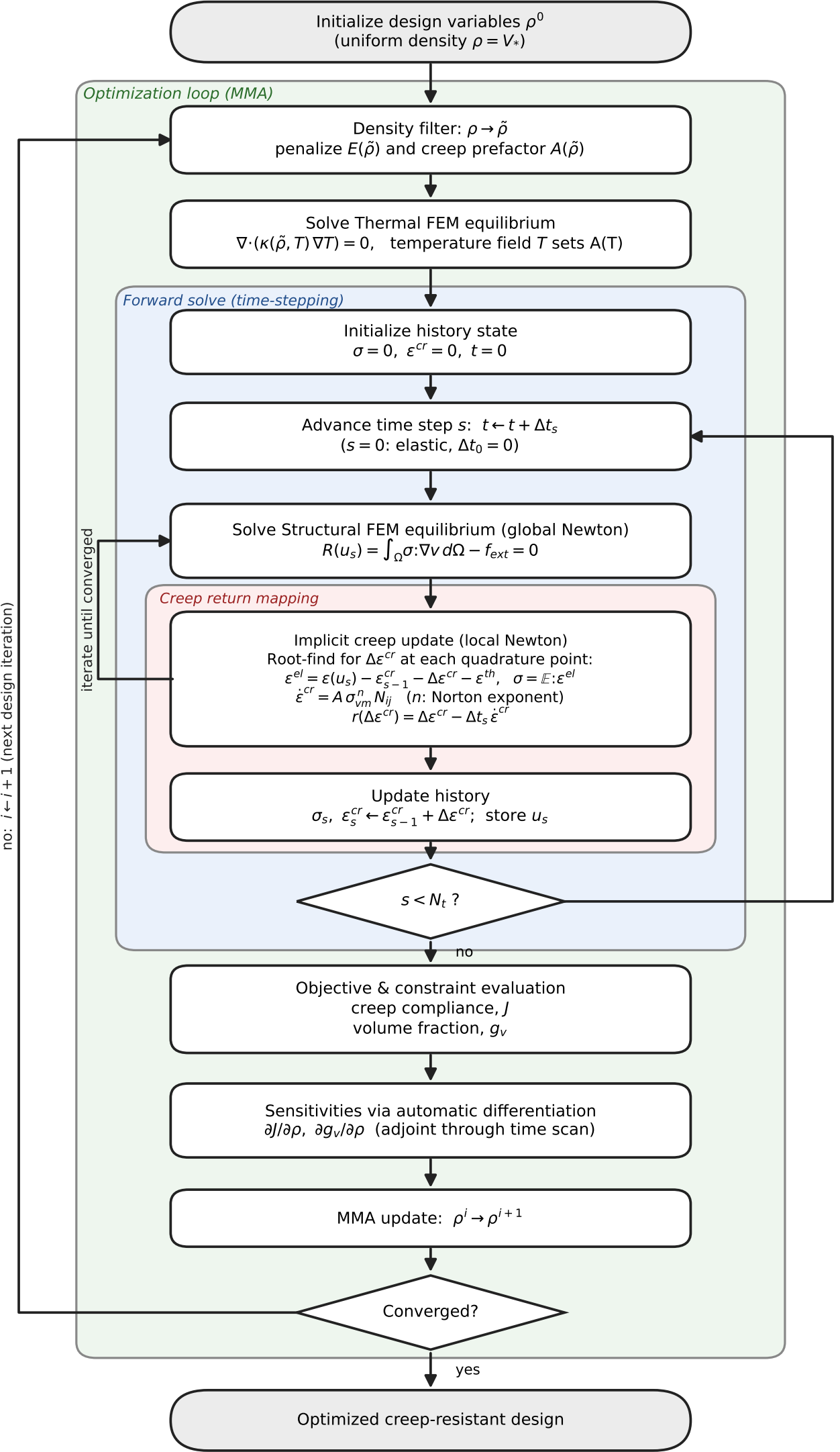
Initialize design variables ρ^0
(uniform density $\rho = V_*$)
Optimization loop (MMA)
Density filter: $\rho \to \tilde{\rho}$
penalize $E(\tilde{\rho})$ and creep prefactor $A(\tilde{\rho})$
Solve Thermal FEM equilibrium
$\nabla\cdot(\kappa(\tilde{\rho}, T)\nabla T) = 0$, temperature field T sets A(T)
Forward solve (time-stepping)
Initialize history state
$\sigma = 0,\ \varepsilon^{cr} = 0,\ t = 0$
Advance time step s: $t \leftarrow t + \Delta t_s$
($s = 0$: elastic, $\Delta t_0 = 0$)
Solve Structural FEM equilibrium (global Newton)
$R(u_s) = \int_\Omega \sigma{:}\nabla v\, d\Omega - f_{ext} = 0$
iterate until converged
Creep return mapping
Implicit creep update (local Newton)
Root-find for $\Delta\varepsilon^{cr}$ at each quadrature point:
$\varepsilon^{el} = \varepsilon(u_s) - \varepsilon^{cr}_{s-1} - \Delta\varepsilon^{cr} - \varepsilon^{th}$, $\sigma = \mathbb{E}{:}\varepsilon^{el}$
$\dot{\varepsilon}^{cr} = A\,\sigma_{vm}^{n}\,N_{ij}$ (n: Norton exponent)
$r(\Delta\varepsilon^{cr}) = \Delta\varepsilon^{cr} - \Delta t_s\,\dot{\varepsilon}^{cr}$
Update history
$\sigma_s,\ \varepsilon^{cr}_s \leftarrow \varepsilon^{cr}_{s-1} + \Delta\varepsilon^{cr}$; store u_s
$s < N_t$?
yes: $s \leftarrow s+1$
no
Objective & constraint evaluation
creep compliance, J
volume fraction, g_v
Sensitivities via automatic differentiation
$\partial J/\partial\rho,\ \partial g_v/\partial\rho$ (adjoint through time scan)
MMA update: $\rho^i \to \rho^{i+1}$
Converged?
no: $i \leftarrow i+1$ (next design iteration)
yes
Optimized creep-resistant design